D20A **Roll No.19**

Experiment No. 4

Aim:
To develop a blockchain-powered web application using Solidity on Remix IDE and integrate a MetaMask wallet for smart contract deployment and transactions on a test network.

Theory:

A blockchain-based web application can interact with smart contracts to store and manage data on a decentralized network. Solidity is used to implement the contract functionality, whereas Remix IDE supports contract development, compilation, deployment, and interaction. MetaMask serves as a browser wallet that enables users to authorize blockchain transactions through their accounts.

A testnet is a blockchain environment designed for development and experimentation without interacting with the main production network. In this experiment, Remix IDE is connected to the Sepolia testnet using MetaMask. The `SimpleStorage` contract stores a public unsigned integer and includes two functions: `setValue` to modify the stored value and `getValue` to retrieve it without changing the blockchain state. Contract deployment and state-changing operations require transaction confirmation through MetaMask.

Steps:
1. Install and open MetaMask and create or import a wallet account.
2. Select the Sepolia testnet and provide test Ether to the wallet.
3. Open Remix IDE and create the Solidity file SimpleStorage.sol.
4. Enter the SimpleStorage contract and compile it with a compatible Solidity compiler.
5. Select Browser Extension as the Remix deployment environment and connect the MetaMask account.
6. Select the Sepolia testnet and review the deployment request in MetaMask.
7. Confirm the transaction in MetaMask and deploy the smart contract.
8. Inspect the deployed contract and transaction information shown by Remix and the explorer link.
9. Interact with setValue to submit a value and use getValue to read the stored value.

Code:

```
// SPDX-License-Identifier: MIT
pragma solidity ^0.8.20;

contract SimpleStorage {
    uint256 public value;

    function setValue(uint256 _value) public {
        value = _value;
    }

    function getValue() public view returns (uint256) {
        return value;
    }
}
```

Listing 1. SimpleStorage.sol

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Output:

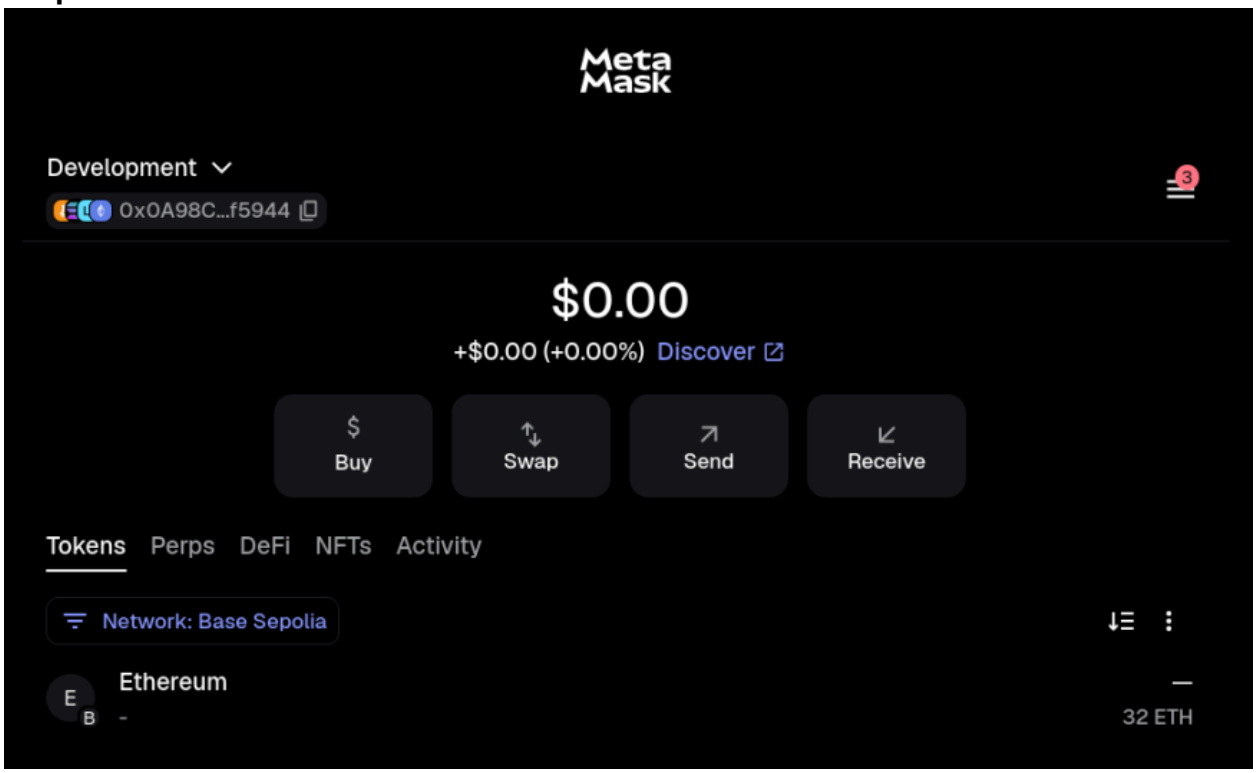


Fig. 1. MetaMask wallet connected to the Base Sepolia network

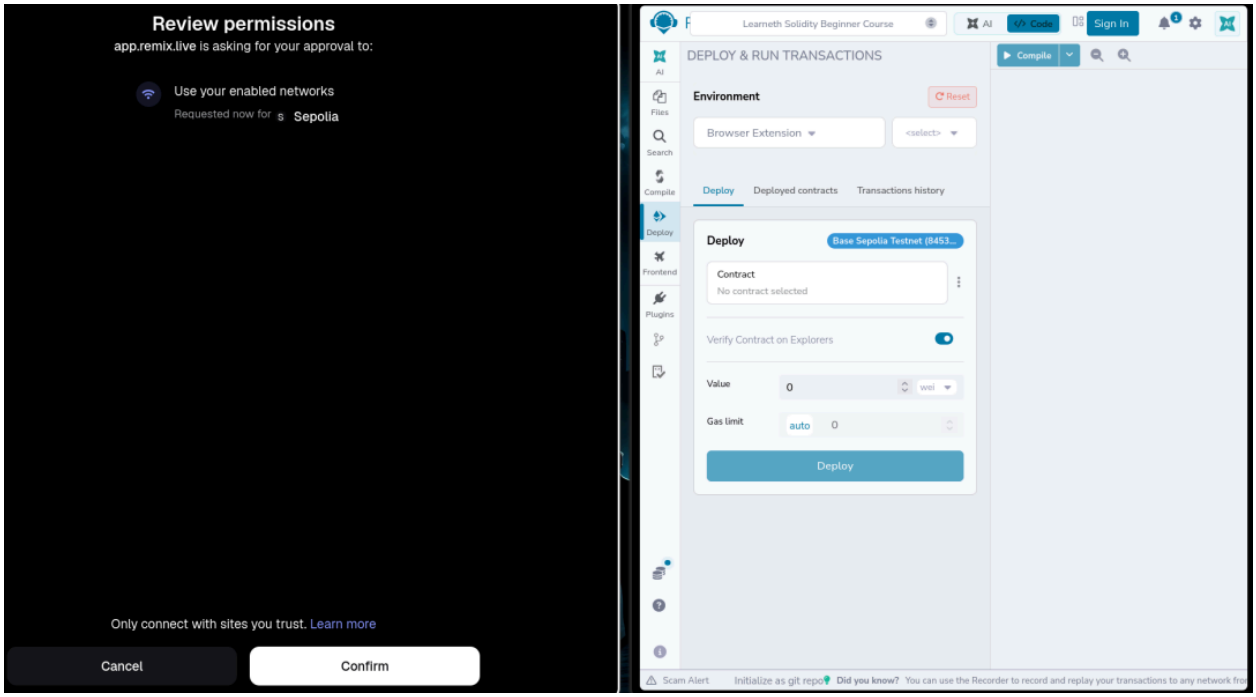


Fig. 2. MetaMask permission request and Remix deployment interface

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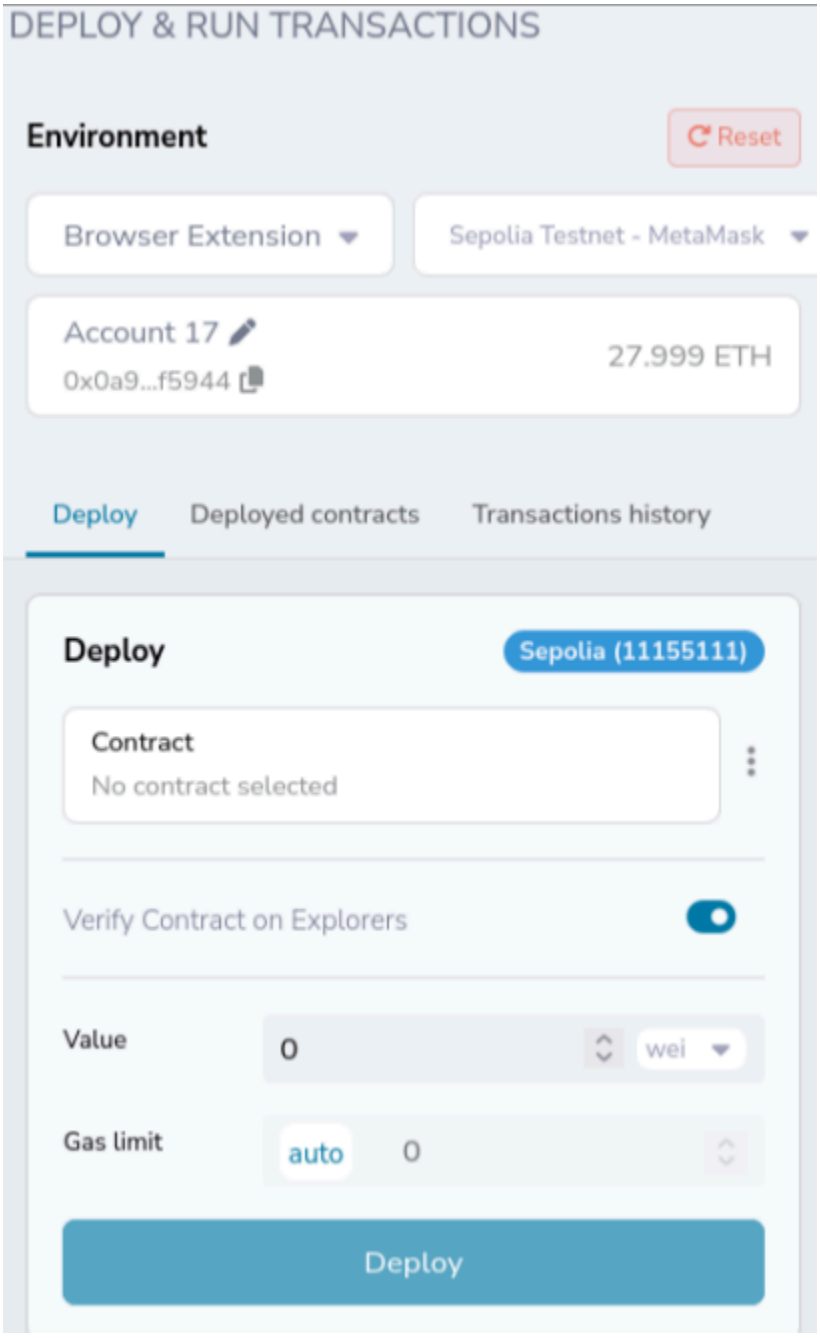


Fig. 3. Remix deployment panel configured for the Sepolia testnet

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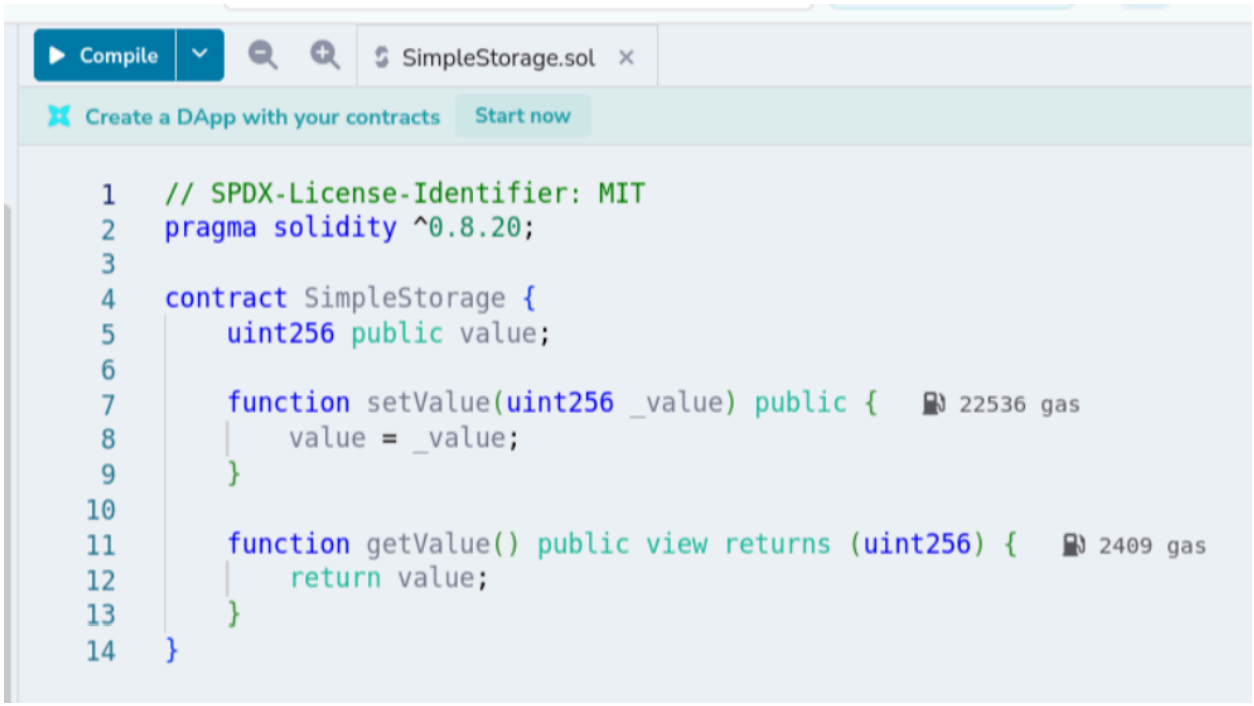


Fig. 4. SimpleStorage contract source code in Remix IDE

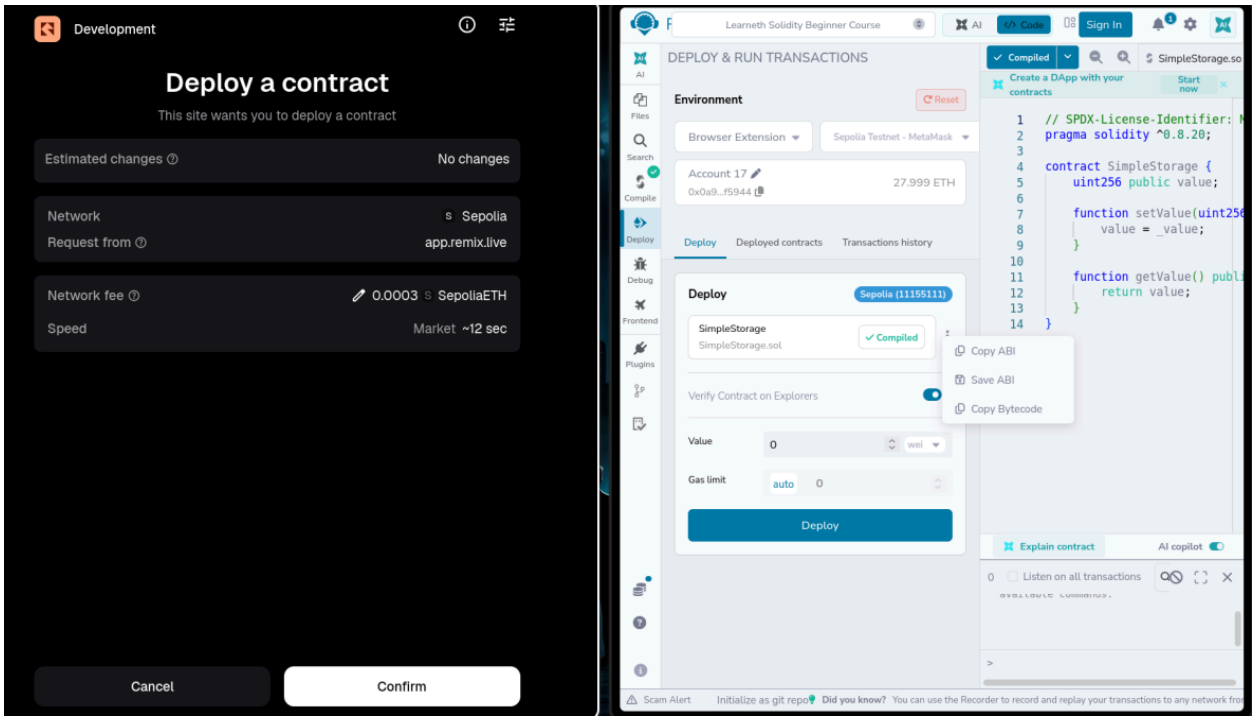


Fig. 5. MetaMask confirmation for deploying the smart contract

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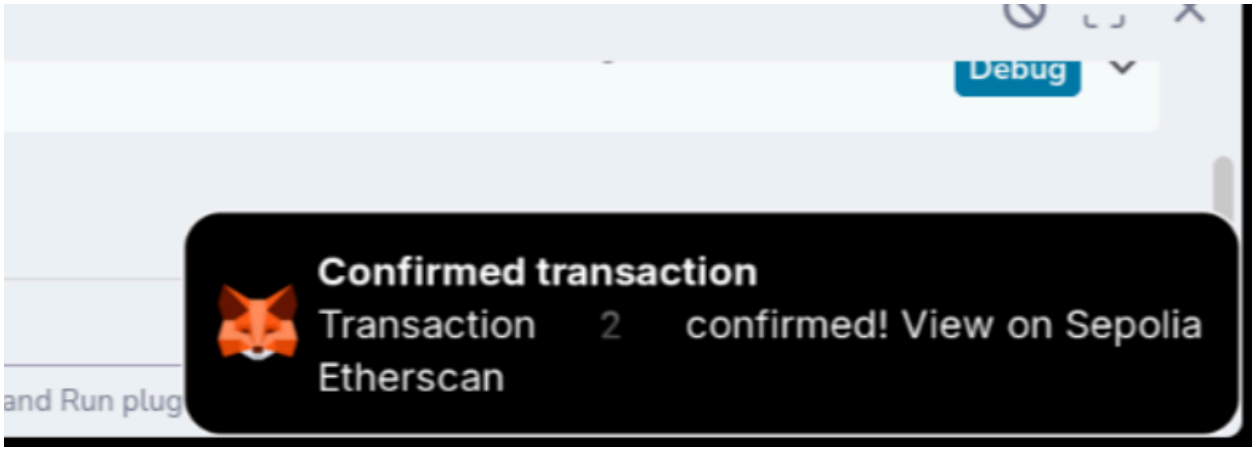


Fig. 6. Deployed contract and confirmed transaction details in Remix

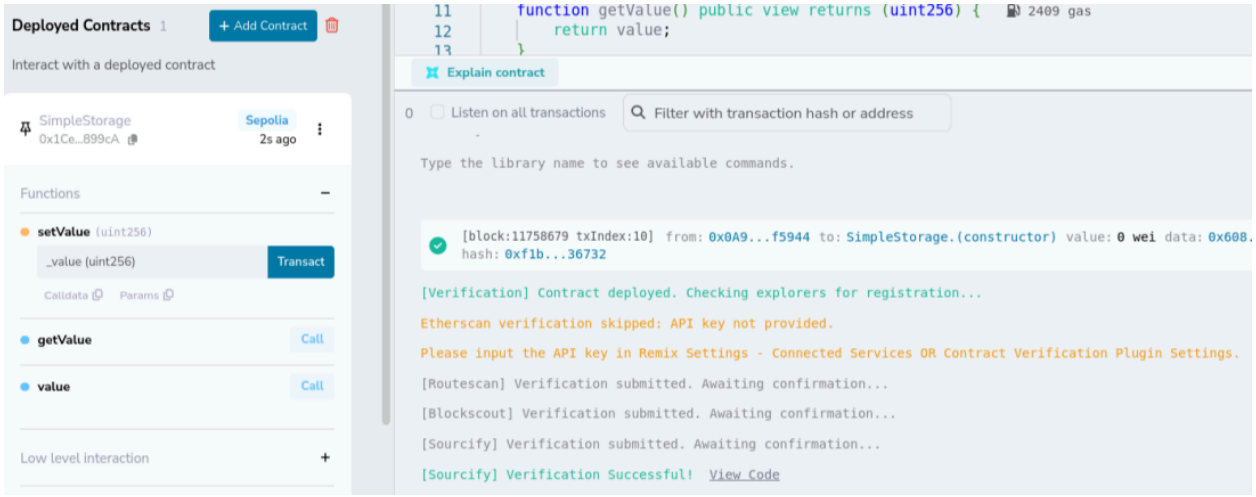


Fig. 7. MetaMask transaction request for a contract interaction

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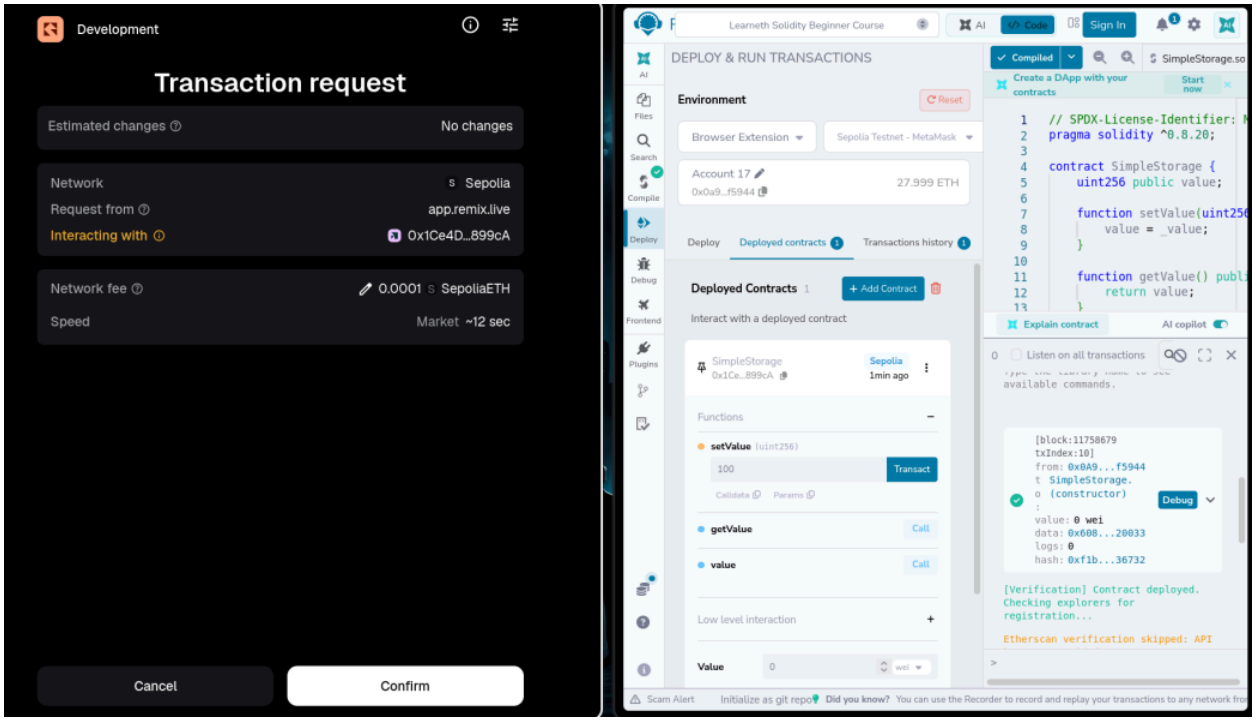


Fig. 8. Remix interface showing the deployed contract interaction

Conclusion:

The `SimpleStorage` smart contract was developed using Solidity and implemented through Remix IDE. The provided screenshots illustrated MetaMask connected to a test network, Remix configured for browser-wallet deployment, deployment authorization, successful transaction confirmation, and subsequent contract interaction. This experiment demonstrated the process of integrating MetaMask with Remix IDE and executing smart contract operations on a blockchain testnet.